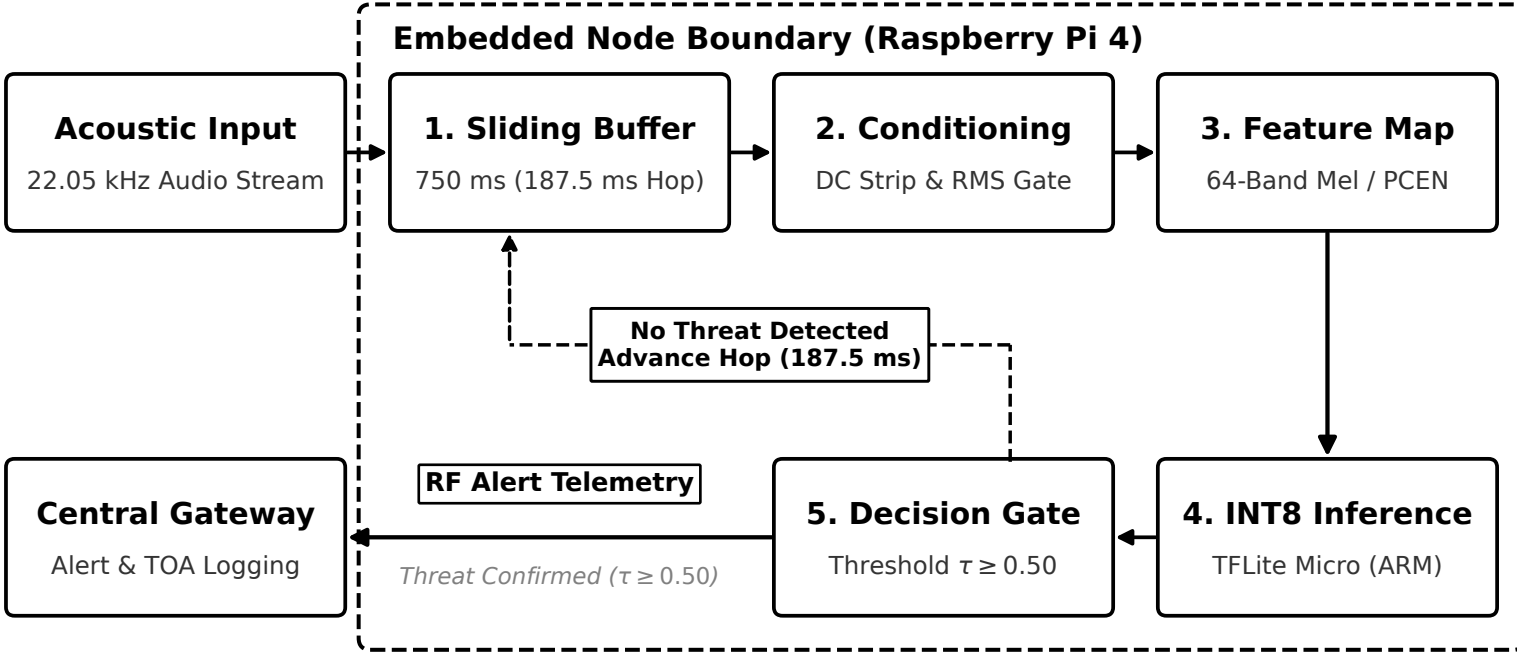
Embedded Node Boundary (Raspberry Pi 4)
Acoustic Input
22.05 kHz Audio Stream
1. Sliding Buffer
750 ms (187.5 ms Hop)
2. Conditioning
DC Strip & RMS Gate
3. Feature Map
64-Band Mel / PCEN
No Threat Detected
Advance Hop (187.5 ms)
4. INT8 Inference
TFLite Micro (ARM)
5. Decision Gate
Threshold $\tau \geq 0.50$
RF Alert Telemetry
Threat Confirmed ($\tau \geq 0.50$)
Central Gateway
Alert & TOA Logging